

RESEARCH ARTICLE | APRIL 08 2025

Visualizing digital twins of fusion power plants using NVIDIA Omniverse

Nitesh Bhatia ; Rui Costa; Stanislas J. P. Pamela ; Andrew Davis; Alejandra N. Gonzalez Beltran; Vignesh Gopakumar; Lorenzo Zanisi ; Shaun de Witt; Robert Akers



AIP Advances 15, 045018 (2025)

<https://doi.org/10.1063/5.0261883>



View
Online



Export
Citation

Articles You May Be Interested In

The impact of design parameters on tritium breeding in a water cooled blanket for CFETR

AIP Advances (November 2025)

Study on neutronics modeling with 22.5° model using ANSYS for CFETR

AIP Advances (June 2023)

Investigate the impact of different neutron source models on the neutronics performance for CFETR

AIP Advances (March 2025)

29 March 2026 19:39:12

AIP Advances

Why Publish With Us?



19 DAYS
average time
to 1st decision



500+ VIEWS
per article (average)



INCLUSIVE
scope

[Learn More](#)

Visualizing digital twins of fusion power plants using NVIDIA Omniverse

Cite as: AIP Advances 15, 045018 (2025); doi: 10.1063/5.0261883

Submitted: 31 January 2025 • Accepted: 10 March 2025 •

Published Online: 8 April 2025



Nitesh Bhatia,^{a)} Rui Costa, Stanislas J. P. Pamela, Andrew Davis, Alejandra N. Gonzalez Beltran, Vignesh Gopakumar, Lorenzo Zanisi, Shaun de Witt, and Robert Akers

AFFILIATIONS

United Kingdom Atomic Energy Authority, Culham Campus, Abingdon OX143DB, United Kingdom

^{a)} Author to whom correspondence should be addressed: nitesh.bhatia@ukaea.uk

ABSTRACT

This paper discusses the NVIDIA Omniverse platform for visualizing and integrating digital twins of fusion power plants, focusing on enhancing the design, simulation, and operational workflows of complex fusion systems. As global fusion projects such as ITER, STEP, and CFETR move toward realizing practical fusion energy, sophisticated tools to model, simulate, and optimize such systems become increasingly critical. Digital twins are virtual replicas of physical systems that act as a “single source of truth” by integrating scientific analysis, engineering design, and real-time data integration. This paper outlines the workflow for creating and visualizing a digital twin of a tokamak, demonstrating the integration of computer-aided design models, simulation data, and material properties into Omniverse. It also provides a view of extending Omniverse with real-time data visualization, Python scripting, and generative AI in applications that make the digital twin even more functional and interactive, allowing seamless collaboration across teams and stakeholders. The interoperability challenges that limit the adoption of Omniverse in fusion research are also discussed. In conclusion, the paper outlines how Omniverse can create a comprehensive, immersive, and interactive environment that optimizes the design and operation of a fusion power plant. In the future, this approach may extend to developing more efficient and reliable fusion energy systems.

© 2025 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>). <https://doi.org/10.1063/5.0261883>

I. INTRODUCTION

Nuclear fusion—the type of energy that is found in the sun—is one of the most promising energy sources that could become sustainable and almost unlimited.^{1–3} Currently, there are ongoing international projects aiming to leverage this vast energy potential, such as ITER (International Thermonuclear Experimental Reactor) in France, STEP (Spherical Tokamak for Energy Production) in the UK, and CFETR (China Fusion Engineering Test Reactor, which aims to build on the operational experimental tokamaks and commercial fusion reactors) in China.⁴ ITER’s objective is to demonstrate fusion energy’s scientific and technical feasibility at an unprecedented scale and, therefore, help in the development of commercial fusion power plants.⁵ STEP aims at establishing the world’s first prototype fusion power plant in the UK and concentrating on innovations in compact tokamak design.⁶ The CFETR intends to bridge the gap between experimental tokamaks and commercial fusion power plants and focuses on improvements made

in long-pulse operation and power generation.⁷ Collectively, these projects represent the next-generation stepping stones to addressing the world’s future energy provision.⁸

The path toward achieving fusion energy, however, does not come without real challenges.^{8,9} Fusion power plants will likely be some of the most complex engineering systems ever manufactured.^{1,10} They will demand the integration of many disciplines, including plasma physics, materials science, thermal hydraulics, and control systems.¹¹ The complex interactions of these systems with stringent safety and performance requirements pose significant obstacles for researchers and engineers.^{12,13} The design, operation, or maintenance of such facilities requires interdependent components, plasma-facing first wall materials, superconducting magnets, power systems, and diagnostics—all of which must work together under conditions of extreme performance.^{14,15}

Digital twins have increasingly played a transformative role in addressing all of these aspects excellently, serving as the single source of truth that ties scientific analysis to engineering workflows for a

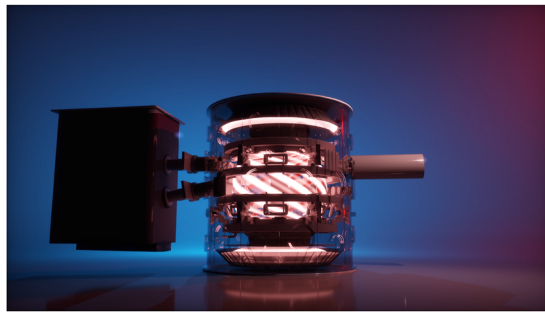


FIG. 1. Visualization of MAST-U tokamak with transparent walls: Rendered using NVIDIA Omniverse, this image showcases the MAST-U tokamak, with its walls represented by a transparent material to reveal the internal components. The glowing plasma filaments inside are generated using JOEREK MHD simulation data (see Sec. III C), visualizing phenomena such as disruptions or instabilities (multimedia available online).

holistic representation of the fusion systems.^{16,17} By connecting real-time data, simulations, and prediction-based models, digital twins replicate the physical plant in a dynamic virtual counterpart, providing resources for design optimization, performance monitoring, and predictive maintenance.^{18,19}

Recent technological advances have vastly widened the efforts regarding digital twins.^{20–23} Data science developments encompass machine learning and present higher quality and scalable analysis of fusion systems. In addition, exascale computing allows running simulations at unprecedented speed for increasingly detailed models.^{24,25} Photorealistic rendering as seen in Fig. 1 (multimedia available online) and immersive virtual reality are a few examples of advanced visualization techniques that create easy intuitive access to complex data for better collaboration and decision-making.^{26–29} All these innovations bring the future of fusion energy close to realizing the vision of commercially viable fusion power plants.³⁰

This paper provides a concise overview of visualizing digital twins using Omniverse in fusion power plant engineering. Section II covers an overview of digital twins and their integration with advanced platforms like NVIDIA Omniverse. Section III presents a detailed workflow for visualizing a digital twin of the MAST-U tokamak, emphasizing the integration of CAD, the assignment of PBR materials, and the incorporation of simulation data. Section IV discusses how NVIDIA Omniverse can be enhanced through Python scripting and the development of custom applications specifically designed to address challenges in fusion research. Section V discusses possible challenges hindering the adoption of Omniverse in fusion research. Section VI concludes the paper by summarizing important findings, their implications in visualization, and identifying potential future directions leveraging Omniverse for improving digital twins' role in fusion research.

II. BACKGROUND

Digital twins are one of the foundational pillars of modern engineering and scientific applications.^{31–34} They provide a framework for system design, analysis, and operation.³⁵ The digital twin of a physical system is a dynamic virtual model consisting of real-time data, simulations, and predictive analytics to offer a representation

of system behaviors.³⁶ This concept has been widely adopted in diverse fields, such as design, manufacturing, aerospace, automotive, and energy.^{37,38} The scope of systems engineering has been expanded by digital twins, enabling engineering tests, design optimization, system performance monitoring, and maintenance prediction without disrupting physical operations. In fusion energy, where safety, reliability, and precision are critical, digital twins are transforming how fusion systems are modeled, developed, and maintained.^{39–41}

Current methods for modeling fusion systems rely primarily on stand-alone simulations and engineering tools focused on individual components or processes, such as plasma behavior, thermal loads, or structural integrity.^{42,43} While these methods have provided valuable insights, they are typically implemented as stand-alone solutions, disconnected from other subsystems and real-world data streams. This compartmentalization hinders the ability to understand the intricate interdependencies within a fusion power plant. Traditional modeling approaches lack the scalability and real-time responsiveness necessary for large-scale projects such as ITER, STEP, or CFETR.^{8,44} There have been attempts to develop coupled surrogate models that integrate various simulation tools with real-time effectiveness, but they have not been scaled to complex scenarios.^{45–48} These limitations underscore the need for an integrated and adaptive framework, a gap that digital twins are uniquely positioned to fill.

NVIDIA Omniverse is a digital tool enabling access to digital twins in fusion engineering on a single platform for integration, visualization, and collaboration.^{49,50} Omniverse is based on Universal Scene Description (USD), which is an open and extensible ecosystem for 3D worlds—and can access CAD models, simulation data, and up-to-date real-time data in a unified environment for the management of digital twins.⁵¹ Omniverse Kit software enables real-time simulation and rendering of 3D environments, while Omniverse Apps, such as Create, View, and Code, support content creation, interactive visualization, and extension development.⁵² Next, Omniverse Connectors offer data interchange compatibility with some of the design and engineering software, such as Autodesk Alias, Trimble SketchUp, etc.⁵³ Omniverse Nucleus is a central file server, allowing multi-user access to digital assets and synchronized updates to the USD files.⁵⁴ These components support importing large datasets of CAD designs and simulations as USD files, maintaining format consistency across design iterations. Through NVIDIA GPU supported ray-traced and path-traced rendering integration, Omniverse enhances the visualization of complex geometries, material properties, and overlaid datasets, supporting engineering analysis and decision-making.⁵⁵

III. DIGITAL TWIN WORKFLOW FOR FUSION SYSTEMS

Within fusion research, engineering designs of tokamaks or their components, such as divertors or blanket modules, can be imported into NVIDIA Omniverse as USD files, ensuring native format support and consistency across design iterations. These models can be enhanced with Physically Based Rendering (PBR) materials, improving realism for stakeholder communication and decision-making. More importantly, complex geometries and specific data types benefit significantly from path-traced and ray-traced rendering, which aids in depth perception, realistic lighting, soft

shadows, and global illumination, helping to distinguish intricate components.^{56–58} Transparency and refraction effects enable accurate visualization of diagnostic windows, coolant channels, and plasma-facing materials, while improved material fidelity ensures that metallic, dielectric, and emissive surfaces appear true to life. In addition, ray tracing aids in large-scale data visualization by clearly separating overlapping datasets, such as thermal or stress maps overlaid on CAD models.⁵⁹ Omniverse enables the integration of scientific simulations, such as plasma dynamics from JOEKE or structural analyses using finite element models like MOOSE, into a unified digital twin framework.^{60,61} By developing custom connectors or NVIDIA connectors with tools like ParaView, simulation outputs can be visualized and merged with real-time CAD models into a single source of truth, bridging scientific analysis with engineering workflows.⁶² The integration supports real-time collaboration through Omniverse Nucleus, enabling researchers, engineers, and operators to interact with the digital twin in an iterative design process. In a nutshell, the design of a digital twin for fusion systems involves integration across several layers of modeling, simulations, and data to fully capture the complexity of the power plant and its components. The following workflow outlines the significant steps involved in developing a digital twin for the MAST-U tokamak using NVIDIA Omniverse.

A. CAD data integration

The initial step in building the digital twin is integrating the CAD model of the power plant and its components into the virtual environment. For MAST-U, the CAD model is retrieved from a central asset repository, providing the main database for engineering designs. This ensures that the most up-to-date design model of MAST-U is used. Then, the model is imported into Omniverse through CAD applications such as SketchUp and Omniverse Connectors for CAD files, supporting several formats and providing flexibility and manageability (Fig. 2).

B. PBR material assignment

After the CAD model is imported into Omniverse, the next stage involves assigning realistic materials onto it using Omniverse USD Composer, a 3D scene authoring application that allows selecting PBR material either via NVIDIA's repository or using Adobe Substance. PBR materials are essential to simulate the way light interacts with surfaces more realistically, establishing an accurate visual model with appropriate representations of lighting effects such as reflections, roughness, and metallicity.⁶³ In doing this for the MAST-U reactor, standard materials from the Omniverse Material Library are applied as the base for most of the component textures. Furthermore, there is an ongoing collaboration between the University of Manchester and UKAEA to review the photorealistic material properties of the fusion components. This research defines the properties of the Bidirectional Reflectance Distribution Function (BRDF) for material emissivity, roughness, and normal maps to obtain accurate textures and lighting effects for components.⁶⁴ The final PBR materials were then placed into Omniverse, adding to the model's realism and making it apt for design reviews and decision-making.

C. Simulation data integration

The model is enhanced by embedding additional simulation data, providing insights into plasma behavior, visible light emission, and dynamics within the tokamak. For instance, in Fig. 3, the JOEKE MHD simulation data depicting experimental plasma behavior through light emission from Edge Localized Modes (ELMs) have been post-processed using Kitware's ParaView.⁶⁵ JOEKE, a magnetohydrodynamics (MHD) plasma code, simulates complex phenomena, including ELMs and the resulting heat loads on plasma-facing components. The simulation data files were processed at the Cambridge Service for Data Driven Discovery (CSD3) High-Performance Computing (HPC) cluster and exported into the Omniverse environment via the Omniverse ParaView Connector. This connector enables real-time visualization of simulation data

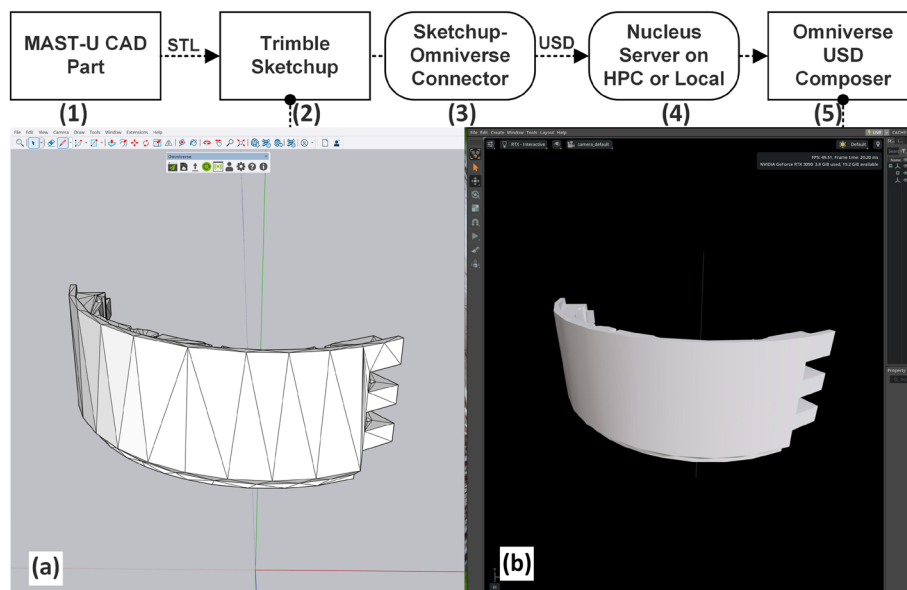


FIG. 2. Workflow for visualizing CAD components from the CAD model of MAST-U in NVIDIA Omniverse. (1) CAD data access and sharing (STL File): CAD data representing engineering designs is obtained using UKAEA's central design repository. (2) The STL CAD file is imported into the SketchUp CAD application for preprocessing refinement of any imperfections or defects. (a) Image supporting SketchUp visualization. (3) SketchUp Omniverse connector: The processed data are written to USD format using the SketchUp Omniverse connector. (4) Nucleus file server: The USD files are hosted in the Omniverse Nucleus server. (5) Omniverse USD Composer: The USD files are visualized using Omniverse USD Composer. (b) Image supporting visualization in USD Composer.

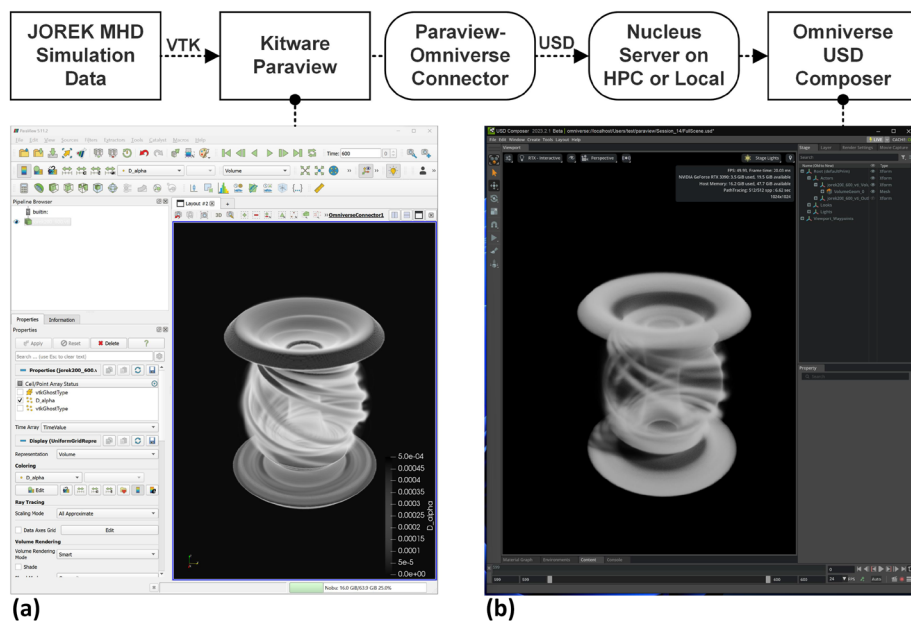


FIG. 3. Workflow for visualizing simulation data from the JOREK MHD simulation in NVIDIA Omniverse. (1) JOREK MHD simulation data sharing (VTK format): Simulation data depicting plasma phenomena such as Edge Localized Modes (ELMs) and D_α light emission are generated and exported in VTK format. (2) Kitware ParaView: The VTK data are processed and visualized in ParaView, enabling a detailed analysis of plasma behavior. (3) ParaView Omniverse connector: The processed data are written to USD format using the Omniverse ParaView connector. (4) Nucleus file server: The USD files are hosted in the Omniverse Nucleus server. (5) Omniverse USD Composer: The USD files are visualized using Omniverse USD Composer. (b) Image supporting visualization in USD Composer.

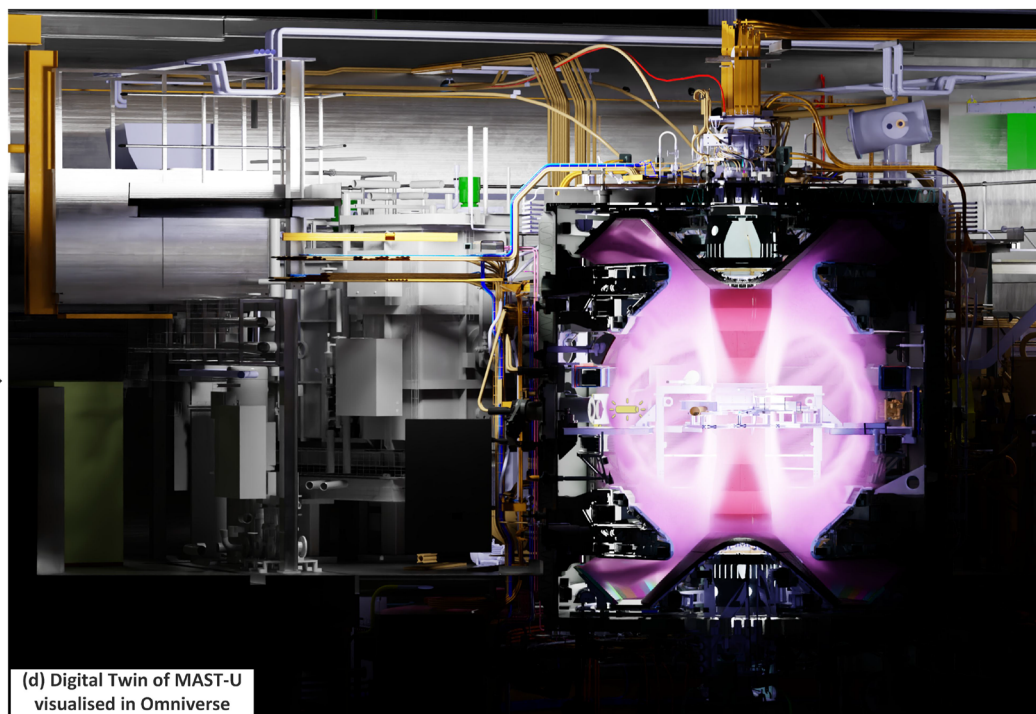
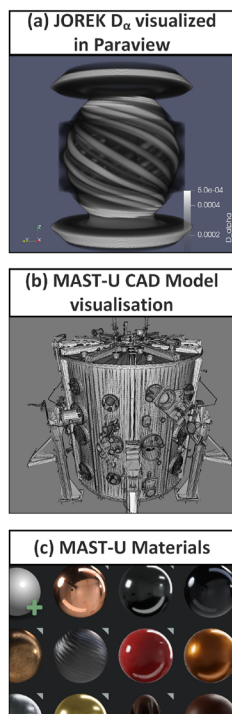


FIG. 4. Visualization of a digital twin of the MAST-U tokamak using NVIDIA Omniverse. (a) Simulation data (JOREK MHD simulation of edge-localized mode plasma in D_α mode), (b) CAD model of the tokamak device, and (c) photorealistic materials are integrated in NVIDIA Omniverse. (d) Cross-sectional rendering of the integrated USD model, showcasing realistic materials and volume rendering of the simulated plasma.

in ParaView, which is streamed live into Omniverse Nucleus as a USD file. Integrating simulation data with the CAD model enables dynamic and interactive digital twin creation, which not only reflects the static design but also visualizes system operation and simulated plasma behavior during operation.

D. Visualizing a unified digital twin of MAST-U tokamak

The consolidation of the CAD model and simulation data into a digital twin is the last step of the activity. Using Omniverse USD, the final step in the workflow is the integration of both the CAD model and simulation data into a cohesive digital twin. Using Omniverse USD Composer, the CAD USD model and the simulation USD model are combined into a unified scene, creating a “single source of truth” representing both the physical design and the dynamically simulated plasma behavior. Integrated modeling is crucial for successfully representing all components of the fusion system, including material properties and behavior in plasma, all within a cohesive environment.¹⁷ Visualization methods like lighting, camera alterations, and animation even further strengthen the real-world feel of the model and, therefore, make it more amenable to vetting in virtual-reality walk-throughs and serious design reviews. For example, the volume visualization of the “ $D\alpha$ ” variable in one of the time steps obtained from an ELM simulation, shown in Fig. 4, helps to illustrate the plasma behavior at that moment in time. Ambient and directional lighting are adjusted to simulate real-world lighting conditions, enhancing the visual fidelity of the scene. Using the final USD file, the stakeholders can interact in real time with the digital twin, perform virtual walk-throughs, and jointly create informed decisions about the design, operation, and maintenance of the fusion power plant. This collaborative aspect is a key advantage of using Omniverse, as it enables teams to engage with the model from various locations, ensuring that all perspectives are considered in the decision-making process.²⁶

IV. EXTENDING OMNIVERSE WITH PYTHON AND CUSTOM SCRIPTS

In order to fully harness the potential of digital twins for fusion systems, functionality should go beyond the standard tools. NVIDIA Omniverse supports Python scripting and can also be turned into customized applications using the Omniverse Kit API that links the digital twin to external systems, provides workflow control, and facilitates intricate simulations.⁶⁶ This allows external real-world operations from robotics to manufacturing facilities and Building Information Modeling (BIM) systems to easily integrate into typical shop-floor operational environments.

A. Real-time interaction with robots and manufacturing systems

Custom Omniverse scripts for interfacing with robotics and manufacturing systems can be used for real-time communication and synchronization. For example, modeling and controlling robotic arms that perform maintenance or component assembly tasks within fusion facilities could be performed in Omniverse. These models are driven by Python scripts that allow the simulation of robotic paths, collision detection, and execution of tasks, all

before being applied to real robots. Error and workflow efficiency are highly minimized in these large-scale production environments, especially within the context of high-precision industries, including energy, automotive, and aerospace sectors.¹⁷

B. Connecting with BIM systems

Fusion power plants constitute major, multifaceted infrastructure typically designed and operated using BIM systems. Python, C++, or web-based custom connectors can interlink Omniverse digital twins with BIM platforms to closely synchronize architectural designs, construction workflows, and maintenance plans with the engineering models of power plants. This covers the entire view of the facility, allowing one to conceive how any changing structure will affect the system as a whole.⁶⁷

C. Simulations and AI integration

Python scripting within the Omniverse environment serves to execute and integrate all advanced simulations directly into the platform. For example, running and viewing simulations on plasma dynamics, irradiation effects, or even thermal loads in present-day reality is possible. Artificial intelligence models, such as surrogate models that predict component wear or optimize operational parameters, can also be integrated into the digital twin through Python APIs.^{68,69} This keeps the digital twin dynamic instead of being a static representation on the grounds that it may help predictive maintenance, design optimization, and operational planning.^{24,70}

D. Real-world data streams

The real-time deployment of sensor data from an actual power plant can be enabled through a digital twin using Python scripting and networked data streams similar to those used for JET and MAST-U diagnostics.^{71,72} Message queues such as MQTT facilitate efficient sensor data streaming from the tokamak and power plant components into the Omniverse environment, supporting real-time digital transformation.¹⁶ The virtual sensors created in the Omniverse and Isaac platform can simulate and validate behavior in real sensors and data-streams.⁷³ Integrating such sensors within a digital twin can have several advantages such as monitoring key variables, detecting anomalies, identifying potential design issues, and testing corrective actions in virtual environments prior to their application in the real world.

V. CHALLENGES

As explained in Secs. III and IV, NVIDIA Omniverse is the ecosystem for enabling real-time 3D simulation and design collaboration. However, its current architecture presents several challenges for interoperability, particularly when bridging the diverse needs of HPC, AI research, and design visualization. Some challenges that hinder adoption and cross-platform integration are discussed in this section. These challenges may change as Omniverse continues to evolve.

Currently, Microsoft Windows systems dominate design and manufacturing industries where CAD and CAE tools like CATIA, SolidWorks, and 3DS Max and design tools by Adobe and Autodesk are standard.^{74–76} Omniverse integrates seamlessly with these tools,

leveraging its RTX GPUs and Windows-optimized workflows to deliver high-performance real-time rendering and visualization capabilities. Most of the existing Omniverse connectors are for Windows applications, including those for very popular tools like the USD distribution of Blender. This has enabled Omniverse to thrive in these domains. However, the reliance on Windows-centric infrastructure limits cross-platform compatibility, particularly with Linux-based scientific workflows.

Like other scientific and engineering domains, the fusion research community prominently relies on Linux-based HPC systems.⁷⁷ Despite its availability on Linux, Omniverse lacks feature parity with the Windows version, restricting its functionality and appeal to Linux-based researchers. SLURM batch computing environments are predominantly used in Linux-HPC environments, where compute nodes often lack Internet connectivity due to security and performance constraints.⁷⁸ Omniverse, however, requires an active Internet connection to connect with the Nucleus file server and its material and scene repository for accessing USD files. While deploying a containerized or cloud version of Omniverse is possible, it breaks compatibility with simulation environments that rely on SLURM batch computing environments, further complicating its integration into Linux-based HPC workflows. In addition, the lack of native Omniverse connectors for scientific simulation tools implies that researchers often need to create their own, which hinders adoption and slows integration into scientific workflows. The disparity in capabilities further complicates collaboration between Windows-based design teams and Linux-focused scientific teams, creating inefficiencies when integrating outputs for projects like ITER or STEP.

Moreover, Omniverse Cloud, which relies primarily on the NVIDIA NGC platform, has limited public providers, such as Microsoft Azure and Amazon AWS.⁷⁹ It does not always comply with fusion research facilities' stringent data residency and security requirements. On-premises deployment of Omniverse could help address some of these concerns; however, it would require a significant investment in GPU infrastructure, such as the NVIDIA OVX platform, which can be entirely out of reach for budget-limited research institutions.⁸⁰

In addition, Omniverse's AR/VR capabilities, which rely exclusively on NVIDIA CloudXR supported on Windows, pose another challenge.⁸¹ The fusion industry is increasingly interested in leveraging AR/VR technologies for virtual inspections, design explorations, training, and control room visualizations. The lack of Linux support for these capabilities limits their adoption and cross-platform usability in research facilities that predominantly operate in Linux environments.

VI. CONCLUSION AND FUTURE DIRECTIONS

The development of digital twins for fusion systems is a significant step toward addressing the complex design, operation, and maintenance issues of next-generation fusion power plants. Integrating multiple simulation tools and data sources into one framework, here demonstrated using NVIDIA Omniverse, gives a unified control environment for managing the interplay of physical, computational, and engineering elements at play. Therefore, ensuring synchronized designs, simulations, and operational workflows that promote a more efficient and reliable decision-making process

throughout the fusion facility lifecycle. The scalability of Omniverse to leverage exascale computing is critical for handling the immense computational demands of fusion simulations and global collaborations. As fusion projects like ITER, STEP, and CFETR continue to progress, the ability to scale digital twin frameworks ensures that researchers, engineers, and stakeholders worldwide can contribute to and benefit from these advancements.

One promising future work area is the joining or coupling of AI surrogate models into the entire digital twin ecosystem. Such surrogate models could provide faster simulations with sufficient accuracy and would play a significant role in real-time assessment, predictive maintenance, and iterative design processes.^{82,83} Digital twins will become intelligent systems by embedding these insights from AI that would be able to dynamically optimize and proactively solve problems.

Going forward, it will be a natural progression to expand digital twin capabilities to other fusion systems and components. This expansion would include advanced diagnostics, control systems, advanced materials, and possibly others into the digital twin framework, which will then represent an even more powerful representation of the fusion power plant ecosystem. As these capabilities mature, they will be in a position to transform not only fusion research but also wider applications across complex, interdisciplinary engineering domains. Together with the integrated capabilities of advanced visualization, scalable computation frameworks, and modern AI, this new generation of digital and physical twins will soon represent an absolute must in translating the dream into practical, workable fusion energy. These newer innovations within Omniverse hint toward a bright prospect where digital ecosystems will create core backbones not only in collaborating toward innovations within fusion science but also across operational excellence in multiple disciplines.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Nitesh Bhatia: Writing – original draft (lead); Writing – review & editing (lead). **Rui Costa:** Writing – original draft (equal); Writing – review & editing (equal). **Stanislas JP Pamela:** Writing – original draft (equal); Writing – review & editing (equal). **Andrew Davis:** Writing – original draft (equal); Writing – review & editing (equal). **Alejandra N Gonzalez Beltran:** Writing – original draft (equal); Writing – review & editing (equal). **Vignesh Gopakumar:** Writing – original draft (equal); Writing – review & editing (equal). **Lorenzo Zanisi:** Writing – original draft (equal); Writing – review & editing (equal). **Shaun de Witt:** Writing – original draft (equal); Writing – review & editing (equal). **Robert Akers:** Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

REFERENCES

- ¹R. Kembleton, "Nuclear fusion: What of the future?," in *Managing Global Warming* (Elsevier, 2019), pp. 199–220.
- ²G. McCracken and P. Stott, "Plasma-surface interactions in tokamaks," *Nucl. Fusion* **19**, 889 (1979).
- ³A. M. Bradshaw, T. Hamacher, and U. Fischer, "Is nuclear fusion a sustainable energy form?," *Fusion Eng. Des.* **86**, 2770–2773 (2011).
- ⁴J. Li, M. Ni, and Y. Lu, "The Frontier and perspective for tokamak development," *Natl. Sci. Rev.* **6**, 382–383 (2019).
- ⁵K. Ikeda, "Progress in the iter physics basis," *Nucl. Fusion* **47**, E01 (2007).
- ⁶H. Lux, D. Wolff, and J. Foster, "Commercialization of fusion power plants," *IEEE Trans. Plasma Sci.* **50**, 4401–4405 (2022).
- ⁷Y. Song, J. Li, Y. Wan, Y. Liu, X. Wang, B. Wan, P. Fu, P. Weng, S. Wu, X. Duan *et al.*, "Engineering design of the CFETR machine," *Fusion Eng. Des.* **183**, 113247 (2022).
- ⁸J. Wesson, *Tokamaks* (Oxford University Press, 2011).
- ⁹I. Chatzis and M. Barbarino, "What is fusion, and why is it so difficult to achieve?," *IAEA Bull.* **62**, 4–5 (2021).
- ¹⁰C. E. Kessel, J. P. Blanchard, A. Davis, L. El-Guebaly, L. M. Garrison, N. M. Ghoniem, P. W. Humrickhouse, Y. Huang, Y. Katoh, A. Khodak *et al.*, "Overview of the fusion nuclear science facility, a credible break-in step on the path to fusion energy," *Fusion Eng. Des.* **135**, 236–270 (2018).
- ¹¹L. A. El-Guebaly, "Fifty years of magnetic fusion research (1958–2008): Brief historical overview and discussion of future trends," *Energies* **3**, 1067–1086 (2010).
- ¹²A. Fasoli, "Essay: Overcoming the obstacles to a magnetic fusion power plant," *Phys. Rev. Lett.* **130**, 220001 (2023).
- ¹³C. Llewellyn-Smith and D. Ward, "Fusion power," *Eur. Rev.* **13**, 337–359 (2005).
- ¹⁴D. Reiter, "Edge plasma physics overview," *Fusion Technol.* **33**, 247–260 (1998).
- ¹⁵P. Bruzzzone, "Superconductivity and fusion energy—The inseparable companions," *Supercond. Sci. Technol.* **28**, 024001 (2014).
- ¹⁶J. Lee, B. Bagheri, and H.-A. Kao, "A cyber-physical systems architecture for industry 4.0," *Computer* **48**, 26–33 (2015).
- ¹⁷Y. Jiang, S. Yin, K. Li, H. Luo, and O. Kaynak, "Industrial applications of digital twins," *Philos. Trans. R. Soc. A* **379**, 20200360 (2021).
- ¹⁸J. Cheng, H. Zhang, F. Tao, and C.-F. Juang, "DT-II: Digital twin enhanced Industrial Internet reference framework towards smart manufacturing," *Robot. Comput.-Integr. Manuf.* **62**, 101881 (2020).
- ¹⁹Z. Liu, N. Meyendorf, and N. Mrad, "The role of data fusion in predictive maintenance using digital twin," *AIP Conf. Proc.* **1949**, 020023 (2018).
- ²⁰R. Zhang, F. Wang, J. Cai, Y. Wang, H. Guo, and J. Zheng, "Digital twin and its applications: A survey," *Int. J. Adv. Des. Manuf. Technol.* **123**, 4123–4136 (2022).
- ²¹S. Mihai, M. Yaqoob, D. V. Hung, W. Davis, P. Towakel, M. Raza, M. Karamanoglu, B. Barn, D. Shetve, R. V. Prasad *et al.*, "Digital twins: A survey on enabling technologies, challenges, trends and future prospects," *IEEE Commun. Surv. Tutorials* **24**, 2255–2291 (2022).
- ²²E. Nica, G. H. Popescu, M. Poliak, T. Klietstik, and O.-M. Sabie, "Digital twin simulation tools, spatial cognition algorithms, and multi-sensor fusion technology in sustainable urban governance networks," *Mathematics* **11**, 1981 (2023).
- ²³N. Bhatia, R. Costa, S. Jackson, N. Cummings, S. Pamela, S. de-Witt, and A. N. Gonzalez-Beltran, "Advanced techniques for fusion data visualisation," *Front. Phys.* **13**, 1569248 (2025).
- ²⁴J. Kates-Harbeck, A. Svyatkovskiy, and W. Tang, "Predicting disruptive instabilities in controlled fusion plasmas through deep learning," *Nature* **568**, 526–531 (2019).
- ²⁵J. Dongarra, "The international exascale software project roadmap," *Int. J. High Perform. Comput. Appl.* **30**, 305–311 (2016).
- ²⁶D. A. Bowman, *Interaction Techniques for Common Tasks in Immersive Virtual Environments* (Georgia Institute of Technology, 1999).
- ²⁷E. H. Korkut and E. Surer, "Visualization in virtual reality: A systematic review," *Virtual Reality* **27**, 1447–1480 (2023).
- ²⁸M. Kraus, K. Klein, J. Fuchs, D. Keim, F. Schreiber, and M. Sedlmair, "The value of immersive visualization," *IEEE Comput. Graphics Appl.* **41**, 125–132 (2021).
- ²⁹Y. Zhao, J. Jiang, Y. Chen, R. Liu, Y. Yang, X. Xue, and S. Chen, "Metaverse: Perspectives from graphics, interactions and visualization," *Visual Inf.* **6**, 56–67 (2022).
- ³⁰D. J. Kropaczek, V. Badalassi, P. K. Jain, P. Ramuhalli, and W. D. Pointer, "Digital twins for nuclear power plants and facilities," in *The Digital Twin* (Springer, 2023), pp. 971–1022.
- ³¹M. Grieves, "Digital twin: Manufacturing excellence through virtual factory replication," White Paper, 2014.
- ³²F. Tao, H. Zhang, A. Liu, and A. Y. C. Nee, "Digital twin in industry: State-of-the-art," *IEEE Trans. Ind. Inf.* **15**, 2405–2415 (2019).
- ³³E. Glaessgen and D. Stargel, "The digital twin paradigm for future NASA and US air force vehicles," in *53rd AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference; 20th AIAA/ASME/AHS Adaptive Structures Conference 14th AIAA* (American Institute of Aeronautics and Astronautics, 2012), p. 1818.
- ³⁴S. K. Khaitan and J. D. McCalley, "Design techniques and applications of cyberphysical systems: A survey," *IEEE Syst. J.* **9**, 350–365 (2014).
- ³⁵Y. Zheng, S. Yang, and H. Cheng, "An application framework of digital twin and its case study," *J. Ambient Intell. Humanized Comput.* **10**, 1141–1153 (2019).
- ³⁶H. Jiang, S. Qin, J. Fu, J. Zhang, and G. Ding, "How to model and implement connections between physical and virtual models for digital twin application," *J. Manuf. Syst.* **58**, 36–51 (2021).
- ³⁷S. Boschert and R. Rosen, "Digital twin—The simulation aspect," in *Mechatronic Futures: Challenges and Solutions for Mechatronic Systems and Their Designers* (Springer, 2016), pp. 59–74.
- ³⁸J. Li, L. Margetts, and M. R. Nabawy, "A digital twin for insect wing aerodynamics," in *AIAA SCITECH 2025 Forum* (American Institute of Aeronautics and Astronautics, 2025), p. 1471.
- ³⁹W. Yu, P. Patros, B. Young, E. Klinac, and T. G. Walmsley, "Energy digital twin technology for industrial energy management: Classification, challenges and future," *Renewable Sustainable Energy Rev.* **161**, 112407 (2022).
- ⁴⁰C. Semeraro, A. Olabi, H. Aljaghoub, A. H. Alami, M. Al Radi, M. Dassisti, and M. A. Abdelkareem, "Digital twin application in energy storage: Trends and challenges," *J. Energy Storage* **58**, 106347 (2023).
- ⁴¹Y. Song, M. Xia, Q. Chen, and F. Chen, "A data-model fusion dispatch strategy for the building energy flexibility based on the digital twin," *Appl. Energy* **332**, 120496 (2023).
- ⁴²G. Federici, R. Kemp, D. Ward, C. Bachmann, T. Franke, S. Gonzalez, C. Lowry *et al.*, "Overview of EU DEMO design and R&D activities," *Fusion Eng. Des.* **89**(7–8), 882–889 (2014).
- ⁴³W. Biel, M. Ariola, I. Bolshakova, K. J. Brunner, M. Cecconello, I. Duran, Th. Franke *et al.*, "Development of a concept and basis for the DEMO diagnostic and control system," *Fusion Eng. Des.* **179**, 113122 (2022).
- ⁴⁴Y. Xie, Y. Zhang, and Y. Zheng, "Digital twin-driven design framework for complex products and systems," *Int. J. Adv. Manuf. Technol.* **107**, 2657–2676 (2020).
- ⁴⁵V. Gopakumar and D. Samaddar, "Image mapping the temporal evolution of edge characteristics in tokamaks using neural networks," *Mach. Learn.: Sci. Technol.* **1**, 015006 (2020).
- ⁴⁶L. Zanisi, A. Ho, J. Barr, T. Madula, J. Citrin, S. Pamela, J. Buchanan, F. Casson, V. Gopakumar, and J. Contributors, "Efficient training sets for surrogate models of tokamak turbulence with active deep ensembles," *Nucl. Fusion* **64**, 036022 (2024).
- ⁴⁷S. Dasbach and S. Wiesen, "Towards fast surrogate models for interpolation of tokamak edge plasmas," *Nucl. Mater. Energy* **34**, 101396 (2023).
- ⁴⁸G. Dong, X. Wei, J. Bao, G. Brochard, Z. Lin, and W. Tang, "Deep learning based surrogate models for first-principles global simulations of fusion plasmas," *Nucl. Fusion* **61**, 126061 (2021).
- ⁴⁹N. Corporation, "NVIDIA Omniverse: The platform for building digital twins," NVIDIA Documentation, 2023.
- ⁵⁰P. A. Studios, "Universal scene description: A framework for interchangeable 3D formats," Pixar Documentation, 2023.
- ⁵¹U. Community, "Introduction to OpenUSD," OpenUSD White Paper (2023).
- ⁵²N. Corporation, "Omniverse developer overview," NVIDIA Documentation, 2024.
- ⁵³N. Corporation, "USD connections overview," NVIDIA Documentation, 2023.

- ⁵⁴N. Corporation, "Omniverse nucleus," NVIDIA Documentation, 2023.
- ⁵⁵N. Corporation, "Omniverse materials and rendering," NVIDIA Documentation, 2023.
- ⁵⁶J. Peddie, "Ray-tracing hardware," in *Ray Tracing: A Tool for All* (Springer, 2019), pp. 129–180.
- ⁵⁷S. G. Parker, H. Friedrich, D. Luebke, K. Morley, J. Bigler, J. Hoberock, D. McAllister, A. Robison, A. Dietrich, G. Humphreys *et al.*, "GPU ray tracing," *Commun. ACM* **56**, 93–101 (2013).
- ⁵⁸H. Ludvigsen and A. C. Elster, "Real-time ray tracing using NVIDIA OptiX," in *Eurographics (Short Papers)* (Eurographics Digital Library, 2010), pp. 65–68.
- ⁵⁹E. J. Nielsen, A. Walden, G. Nastac, L. Wang, W. Jones, M. Lohry, W. K. Anderson, B. Diskin, Y. Liu, C. L. Rumsey *et al.*, "Large-scale computational fluid dynamics simulations of aerospace configurations on the Frontier exascale system," in *AIAA Aviation Forum and ASCEND 2024* (American Institute of Aeronautics and Astronautics, 2024), p. 3866.
- ⁶⁰M. Hoelzl *et al.*, J. Team, "The JOEKE non-linear extended MHD code and applications to large-scale instabilities and their control in magnetically confined fusion plasmas," *Nucl. Fusion* **61**, 065001 (2021).
- ⁶¹P.-C. A. Simon, C. T. Icenhour, M. Shimada, A. D. Lindsay, G. L. Giudicelli, L. H. Harbour, D. R. Gaston *et al.*, "Developing multiphysics, integrated, high-fidelity, massively parallel computational capabilities for fusion applications using MOOSE," Report No. INL/CON-24-76614-Rev000, Idaho National Laboratory (INL), Idaho Falls, ID, 2024.
- ⁶²A. H. Squillacote, J. Ahrens, C. Law, B. Geveci, K. Moreland, and B. King, *The Paraview Guide* (Kitware Clifton Park, NY, 2007), Vol. 366.
- ⁶³J. T. Kajiya, "The rendering equation," *ACM SIGGRAPH Comput. Graphics* **20**, 143–150 (1986).
- ⁶⁴M. Pharr, C. Kolb, and R. Gershbein, "Rendering complex scenes with memory-coherent ray tracing," in *Proceedings of the 24th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '97)* (ACM Press/Addison-Wesley Publishing Co, 1997), pp. 101–108.
- ⁶⁵S. Smith, S. Pamela, A. Fil, M. Hölzl, G. Huijsmans, A. Kirk, D. Moulton, O. Myatra, A. Thornton, H. Wilson *et al.*, "Simulations of edge localised mode instabilities in MAST-U Super-X tokamak plasmas," *Nucl. Fusion* **60**, 066021 (2020).
- ⁶⁶NVIDIA, "NVIDIA omniverse kit," <https://developer.nvidia.com/omniverse-kit>, 2023.
- ⁶⁷S. T. Matarneh, M. Danso-Amoako, S. Al-Bizri, M. Gaterell, and R. Matarneh, "Building information modeling for facilities management: A literature review and future research directions," *J. Build. Eng.* **24**, 100755 (2019).
- ⁶⁸V. Gopakumar, S. Pamela, L. Zanisi, Z. Li, A. Gray, D. Brennand, N. Bhatia, G. Stathopoulos, M. Kusner, M. Peter Deisenroth, A. Anandkumar, and JOEKE Team, and M. Team, "Plasma surrogate modelling using Fourier neural operators," *Nucl. Fusion* **64**, 056025 (2024).
- ⁶⁹R. Gaur, V. Gopakumar, N. Barbour, B. Jang, N. Mandell, I. Abel, W. Dorland, and E. Kolemen, "Spatio-temporal forecasting of plasma turbulence using deep learning," in *APS Division of Plasma Physics Meeting Abstracts, APS Meeting Abstracts* (American Physical Society, 2023), p. JO09.015.
- ⁷⁰W. Smith, A. Barker, Z. Miao, O. Woolland, M. Omer, and L. Margetts, "Automated machine learning workflows for fusion power plant design," in *Engineering Computational Technology (ECT)*, 12 (Civil-Comp Press, Edinburgh, UK, 2024).
- ⁷¹P. Millward, A. Ainsworth, C. J. Caldwell-Nichols, R. Lobel, and C. J. Hancock, "Engineering aspects of jet diagnostic systems," *Fusion Technol.* **11**, 235–252 (1987).
- ⁷²J. Milnes, N. B. Ayed, F. Dhalla, G. Fishpool, J. Hill, I. Katramados, R. Martin, G. Naylor, T. O'Gorman, R. Scannell *et al.*, "Mast upgrade—Construction status," *Fusion Eng. Des.* **96**, 42–47 (2015).
- ⁷³N. Corporation, "Building smarter autonomous machines: NVIDIA announces early access for omniverse sensor RTX," NVIDIA Documentation, 2025.
- ⁷⁴J. Koh, CATIA V5 Design Fundamentals, ONSIA, 2017.
- ⁷⁵G. Page, B. Yorke-Biggs, and S. De-Guido, *IDE Digital Twin White Paper: Harnessing the Digital Twin for Real Competitive Advantage* (Loughborough University, 2022).
- ⁷⁶H. Pozniak, "Could engineers work in the metaverse?," *Eng. Technol.* **17**, 99–106 (2022).
- ⁷⁷B. Guillerminet, I. C. Plasencia, M. Haeefe, F. Iannone, A. Jackson, G. Manduchi, M. Plociennik, E. Sonnendruker, P. Strand, M. Owsiak *et al.*, "High performance computing tools for the integrated tokamak modelling project," *Fusion Eng. Des.* **85**, 388–393 (2010).
- ⁷⁸M. A. Jette and T. Wickberg, "Architecture of the Slurm workload manager," in *Workshop on Job Scheduling Strategies for Parallel Processing* (Springer, 2023), pp. 3–23.
- ⁷⁹N. Corporation, "Nvidia omniverse cloud," NVIDIA Documentation, 2023.
- ⁸⁰N. Corporation, "Omniverse at scale: NVIDIA announces third-generation OVX computing systems to power industrial metaverse applications," NVIDIA Documentation, 2023.
- ⁸¹N. Corporation, "NVIDIA CloudXR suite," NVIDIA Documentation, 2023.
- ⁸²P. Mánek, G. Van Goffrier, V. Gopakumar, N. Nikolaou, J. Shimwell, and I. Waldmann, "Fast regression of the tritium breeding ratio in fusion reactors," *Mach. Learn.: Sci. Technol.* **4**, 015008 (2023).
- ⁸³N. Carey, L. Zanisi, S. Pamela, V. Gopakumar, J. Omotani, J. Buchanan, J. Brandstetter, F. Paischer, G. Galletti, and P. Setinek, "Data efficiency and long-term prediction capabilities for neural operator surrogate models of edge plasma simulations," *arXiv:2502.17386* (2025).